DATA & AI PORTFOLIO

M. Thufail Alwannabil Samas

Data Science | Data Engineering | Data Analysis | Machine Learning | Deep Learning

Mathematics graduate and BNSP-certified Data Scientist with practical experience spanning data analysis, data engineering, and machine learning. The work includes Python and PostgreSQL data pipelines, statistical and time-series analysis, supervised and deep learning models, and dashboards that translate analytical outputs into operational information. This portfolio presents selected case studies across the data lifecycle, from ingestion and quality control to modeling, evaluation, and communication.

01

DWDM Monitoring and Assurance

A data engineering and analysis case study covering telemetry ingestion, quality controls, statistical detection, and dashboard delivery.

02

Rainfall Forecasting with XGBoost and Grid Search

A data science and machine learning study using lagged meteorological features and chronological cross-validation.

03

Rainfall Forecasting with BiLSTM and Grid Search

A data science and deep learning study using seven-day multivariate sequences to estimate next-day rainfall.

04

Rice Price Forecasting with ARIMA and Walk-Forward Validation

A data analysis and statistical forecasting study using stationarity diagnostics and walk-forward validation.

05

Thyroid Cancer Recurrence Classification with XGBoost, Information Gain, and Grid Search

A data science and machine learning classification study using fold-wise feature selection and multi-metric evaluation.

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SELECTED WORK

Portfolio scope and competency coverage

The selected cases cover three complementary areas of professional practice: data engineering for reliable data flow, data analysis for interpreting patterns and operational conditions, and data science and AI for statistical and predictive modeling. Together, they demonstrate work across ingestion, validation, transformation, analysis, modeling, evaluation, and delivery.

01	**DWDM monitoring and assurance analytics** Professional system for converting optical telemetry into structured monitoring and investigation outputs.	DATA ENGINEERING DATA ANALYSIS MONITORING ANALYTICS	Python/PostgreSQL pipeline, data-quality analysis, statistical detectors, REST API outputs, and dashboard.
02	**Rainfall Forecasting with XGBoost and Grid Search** Next-day rainfall regression with lagged weather features and time-aware validation.	DATA SCIENCE MACHINE LEARNING TIME-SERIES FORECASTING	Feature engineering, temporal cross-validation, Grid Search, MAAPE/RMSE, and peer-reviewed publication.
03	**Rainfall Forecasting with BiLSTM and Grid Search** Seven-day multivariate sequence modeling for next-day rainfall forecasting.	DATA SCIENCE DEEP LEARNING TIME-SERIES FORECASTING	TensorFlow/Keras, three-layer BiLSTM architecture, chronological holdout, and model evaluation.
04	**Rice Price Forecasting with ARIMA and Walk-Forward Validation** Interpretable monthly forecasting for medium and premium rice prices.	DATA ANALYSIS STATISTICAL MODELING TIME-SERIES FORECASTING	Trend and stationarity analysis, ACF/PACF, residual diagnostics, walk-forward validation, and MAPE.
05	**Thyroid Cancer Recurrence Classification with XGBoost, Information Gain, and Grid Search** Binary recurrence classification with fold-wise feature selection and multi-metric evaluation.	DATA SCIENCE MACHINE LEARNING CLASSIFICATION	Information Gain, 10-fold cross-validation, Grid Search, F1-score, specificity, and ROC-AUC.

Data analysis

Data-quality assessment, aggregation, trend analysis, statistical diagnostics, interpretation, and dashboard-based reporting.

Data engineering

Python and SQL processing, PostgreSQL pipelines, ingestion, transformation, structured data layers, and REST API outputs.

Data science and AI

Statistical modeling, feature engineering, XGBoost, BiLSTM, validation design, tuning, and multi-metric evaluation.

Communication and delivery

Analytical dashboards, technical documentation, published research, reproducible workflows, and responsible interpretation.

DWDM Optical Monitoring and Assurance Analytics

DATA ENGINEERING | DATA ANALYSIS | MONITORING ANALYTICS

Python | PostgreSQL | REST API | HTML/CSS/JavaScript

~400	5 min	P1-P10
OPTICAL SENSORS	TELEMETRY CADENCE	OPERATIONAL PRIORITIES

OBJECTIVE

Design and implement a project-stage data system that transforms distributed DWDM optical telemetry into structured monitoring and investigation outputs. The work combined data engineering for reliable processing with data analysis for quality assessment, event identification, and operational interpretation.

DATA

The source environment combined five-minute optical power readings, daily summaries, downtime and coverage indicators, and sensor and link metadata. The analysis required consistent handling of variable historical windows, missing observations, bounded gaps, and conflicting telemetry states.

SYSTEM WORKFLOW

PRTG telemetry (signals and metadata) → **Ingestion** (synchronization and loading) → **PostgreSQL** (raw, processed, event layers) → **Preprocessing** (quality checks and gap handling) → **Detection** (failure, change, degradation) → **API and dashboard** (monitoring and investigation)

METHODOLOGICAL DESIGN

- Engineered metadata synchronization, telemetry ingestion, preprocessing, bounded-gap handling, processed-series generation, structured PostgreSQL layers, and API-ready outputs.
- Performed data-quality and failure analysis for missing telemetry, zero coverage, downtime indicators, empty rows, low optical power, and contiguous events.
- Developed statistical change-point and gradual-degradation analysis using median baselines, MAD-based noise estimation, Theil-Sen slope, Mann-Kendall testing, persistence, stability, and step-dominance checks.

RESULT AND INTERPRETATION

The resulting data products included active-failure events, change-point and gradual-degradation findings, investigation charts, and P1-P10 sensor prioritization based on recent signal degradation relative to historical best levels. The pipeline organized raw telemetry into analysis-ready outputs, while the dashboard supported structured operational review. No quantified production impact is claimed because the work remained project-stage.

LIMITATION

The project did not include a completed production-impact study. Detector behavior also depends on telemetry quality, parameter selection, and the operational interpretation of optical signal changes.

EVIDENCE

Confidential professional work. Raw data, source code, infrastructure details, and internal screenshots are excluded. The case study is limited to architecture, analytical logic, and responsibilities that can be described safely.

Rainfall Forecasting with XGBoost and Grid Search

DATA SCIENCE | MACHINE LEARNING | TIME-SERIES FORECASTING

Python | Pandas | Scikit-learn | XGBoost

1 day	10 folds	243
LAG REPRESENTATION	TIMESERIESSPLIT	MODEL CONFIGURATIONS

OBJECTIVE

Estimate next-day rainfall in Sumenep Regency from recent meteorological conditions. The study examined whether a tree-based boosting model could represent nonlinear relationships between weather variables and rainfall while preserving the temporal order of evaluation.

DATA

Daily BMKG observations were transformed into a supervised table using one-day lagged predictors. Inputs included temperature, sunshine duration, mean sea-level pressure, relative humidity, wind speed, evaporation, and historical rainfall.

ANALYTICAL WORKFLOW

Daily observations — weather and rainfall → **Lag features** — previous-day predictors → **Fold scaling** — fit on training history → **Temporal CV** — 10 chronological folds → **XGBoost search** — Grid Search comparison → **Forecast evaluation** — MAAPE and RMSE

METHODOLOGICAL DESIGN

- Used TimeSeriesSplit so each validation period followed its corresponding training history, avoiding the unrealistic mixing of earlier and later observations.
- Estimated Min-Max scaling parameters within each training fold to reduce temporal information leakage.
- Selected the tuned configuration using mean cross-validated MAAPE, with RMSE reported in millimeters for additional interpretation.

RESULT AND INTERPRETATION

The selected configuration achieved a MAAPE of **0.9152** and an RMSE of **11.9566 mm**. The model represented recurring low-to-moderate rainfall behavior more consistently than extreme peaks. The aggregate metrics therefore summarize overall forecasting error but do not establish reliable extreme-rainfall detection.

LIMITATION

The dataset represents one observation area, uses a fixed one-day lag design, and was evaluated using non-nested temporal cross-validation. Generalization to other regions, periods, or extreme events requires additional testing.

EVIDENCE

Peer-reviewed publication: MIND Journal, Vol. 11, No. 1, pp. 30-43, June 2026.
doi.org/10.26760/mindjournal.v11i1.30-43

03 | DATA SCIENCE AND DEEP LEARNING STUDY

Rainfall Forecasting with BiLSTM and Grid Search

DATA SCIENCE | DEEP LEARNING | TIME-SERIES FORECASTING

Python | TensorFlow/Keras | Pandas | NumPy

7 x 5	80:20	3
SEQUENCE SHAPE	CHRONOLOGICAL SPLIT	BILSTM LAYERS

OBJECTIVE

Investigate whether a recurrent neural network could use short-term weather sequences to estimate rainfall for the following day. The study focused on temporal representation and model architecture rather than on treating observations as independent tabular records.

DATA

Each sample contained seven consecutive days of five variables: average temperature, sunshine duration, relative humidity, mean wind speed, and historical rainfall. The corresponding target was rainfall on the next day.

SEQUENCE AND MODEL ARCHITECTURE

- **7-day sequence** — five weather variables →
- **Normalization** — scaled model inputs →
- **BiLSTM layer 1** — returns full sequence →
- **BiLSTM layer 2** — returns full sequence →
- **BiLSTM layer 3** — sequence summary →
- **Dense(1)** — next-day rainfall

Grid Search compared recurrent units, batch sizes, and learning-rate schedules.

METHODOLOGICAL DESIGN

- Constructed multivariate sliding-window sequences from chronologically ordered observations.
- Reserved the final 20% of the series as a later holdout period rather than using a random split.
- Applied Min-Max normalization and compared three-layer stacked BiLSTM configurations through Grid Search, selecting the model with the lowest normalized MAAPE criterion.

RESULT AND INTERPRETATION

The best configuration achieved a MAAPE of **0.8073** and an RMSE of **10.2734 mm**. The result indicates that a seven-day sequence can capture useful short-term temporal information. However, the estimate reflects one chronological holdout and should not be interpreted as a complete assessment of performance across seasons or future periods.

LIMITATION

The sample size and rainfall imbalance constrain model comparison. A stronger evaluation would use repeated temporal splits or an additional out-of-time test period, together with more explicit analysis of high-rainfall events.

EVIDENCE

Reproducibility note: the original BMKG observations are private. The public project uses a synthetic sample with the same column structure so that preprocessing and model logic can be reviewed without redistributing restricted data.

Rice Price Forecasting with ARIMA and Walk-Forward Validation

DATA ANALYSIS | STATISTICAL MODELING | TIME-SERIES FORECASTING

Python | Statsmodels | Pandas | Matplotlib

60	48 + 12	1 step
MONTHLY OBSERVATIONS	INITIAL HISTORY + TEST	WALK-FORWARD HORIZON

OBJECTIVE

Analyze monthly movements in medium and premium rice prices in East Java and develop an interpretable forecasting workflow. The study emphasized trend and stationarity assessment, statistical diagnostics, transparent model selection, and realistic sequential evaluation.

DATA

The project used a public 60-month dataset from the East Java Open Data portal. Medium and premium rice prices were examined as separate monthly series, with the first 80% used as the initial analytical and training history.

STATISTICAL FORECASTING WORKFLOW

Monthly prices — medium and premium → **ADF testing** — stationarity and differencing → **ACF/PACF** — candidate p and q orders → **ARIMA search** — candidate comparison → **Diagnostics** — residual behavior → **Walk-forward test** — one-step forecasts

METHODOLOGICAL DESIGN

- Examined the monthly series and selected differencing order through sequential ADF tests on the initial historical window.
- Used ACF and PACF analysis to identify candidate autoregressive and moving-average structures.
- Evaluated eligible ARIMA configurations with expanding-window walk-forward validation and reviewed convergence, residual behavior, and error metrics before interpretation.

RESULT AND INTERPRETATION

The selected models achieved MAPE values of **2.081%** for medium rice and **1.8174%** for premium rice. Within the evaluated period, the forecasts followed the observed monthly price movements relatively closely. The result is best interpreted as evidence for the selected sample and horizon, not as a guarantee of stability under structural market changes.

LIMITATION

The series contains only 60 monthly observations, limiting the depth of seasonal and structural analysis. ARIMA also assumes that historical dependence remains informative; sudden policy, supply, or market shocks may not be captured.

EVIDENCE

Reproducible project: public monthly rice-price dataset, documented notebook and Python workflow, stationarity outputs, ACF/PACF tables, diagnostics, candidate results, and walk-forward predictions.

05 | DATA SCIENCE AND MACHINE LEARNING STUDY

Thyroid Cancer Recurrence Classification with XGBoost, Information Gain, and Grid Search

DATA SCIENCE | MACHINE LEARNING | CLASSIFICATION

Python | Scikit-learn | XGBoost | Pandas

383	10 folds	6 metrics
PATIENT RECORDS	CROSS-VALIDATION	EVALUATION COVERAGE

OBJECTIVE

Classify thyroid cancer recurrence from clinicopathologic variables while examining whether Information Gain could reduce the encoded feature set without sacrificing predictive performance. The study also compared tuned and default XGBoost configurations.

DATA

The Differentiated Thyroid Cancer Recurrence dataset from the UCI Machine Learning Repository contains 383 records, one numeric variable, several categorical clinical variables, and a binary recurrence outcome.

CLASSIFICATION WORKFLOW

Clinical data audit and preparation → **Encoding** numeric model table → **Information Gain** fold-wise feature ranking → **10-fold CV** training and validation → **XGBoost search** tuned and default models → **Evaluation** classification metrics

METHODOLOGICAL DESIGN

- Encoded clinical variables explicitly and calculated entropy-based Information Gain to examine feature relevance.
- Performed feature selection within each training fold, preventing validation observations from influencing the selected feature set.
- Compared Grid Search configurations and default XGBoost models using accuracy as the selection criterion, while reviewing precision, recall, specificity, F1-score, and ROC-AUC.

RESULT AND INTERPRETATION

The selected configuration achieved **0.9725 accuracy**, **0.9537 F1-score**, **0.9805 specificity**, and **0.9860 ROC-AUC**. The combined metrics indicate strong discrimination within this dataset. They do not establish clinical readiness, and the model should be interpreted as a methodological classification study.

LIMITATION

The dataset is small, results may vary with fold assignment, and Information Gain measures association rather than causality. External validation and clinical-domain review would be required before any practical medical use.

EVIDENCE

Reproducible project: UCI dataset reference, documented preprocessing, Information Gain outputs, fold-specific selected features, Grid Search results, fold metrics, and final model comparisons.

PUBLICATION, CREDENTIAL, AND CONTACT

Academic and professional foundation

Prediksi Curah Hujan di Kabupaten Sumenep Menggunakan Metode Extreme Gradient Boosting (XGBoost) dan Algoritma Grid Search

M. Thufail Alwannabil Samas, Nurissaidah Ulinnuha, Moh. Hafiyusholeh
MIND (Multimedia Artificial Intelligent Networking Database) Journal, Vol. 11, No. 1, pp. 30-43, June 2026
DOI: 10.26760/mindjournal.v11i1.30-43

PROFESSIONAL CERTIFICATION

Data Scientist

Indonesian Professional Certification Authority (BNSP)
Valid September 2024 - September 2027

Competency coverage includes business and technical objective definition, data analysis and validation, cleaning and construction, model development, evaluation, and review.

EDUCATION

Bachelor of Mathematics

UIN Sunan Ampel Surabaya
August 2021 - January 2025
GPA: 3.46/4.00

Completed the degree in seven semesters, with coursework in artificial intelligence, programming, statistics, multivariate analysis, numerical methods, and applied linear algebra.

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> This portfolio complements the resume by presenting evidence of data engineering, data analysis, and data science and AI practice across pipeline implementation, statistical investigation, model development, evaluation, dashboard delivery, and responsible interpretation.